START: match the question to a worked method

Measurements: one sample vs a fixed number → one-sample t; two independent groups → pooled t; several equal-size groups → balanced ANOVA. **Category counts:** two groups/two outcomes → corrected z or 2×2 χ^2; more rows/columns → ordinary contingency χ^2; one list vs stated probabilities → goodness of fit. **After ANOVA:** requested pairs + Bonferroni/Holm. **"How large?":** prospective → RR; case-control → OR; absolute benefit/harm → NNT/NNH.
Probability: a list of possible values → discrete; a density over a range → integrate; Normal individual → use SD; Normal sample average → use SEM. **Given a probability, asked for a cutoff?** Reverse the CDF/Normal lookup.

Keep each number beside its job

X (random variable); x_i (one observed value); $\bar{x}$ (sample mean); μ (population mean); s (sample SD); σ (population SD); s^2, σ^2 (variances); n (subjects); m (groups); ν (degrees of freedom). A bar means average; a hat means estimate; $\sum$ means add. **SD → square → variance. Variance → square root → SD.**

Raw measurements → mean, variance, SD, SEM

Given heights: $65, 69, 59, 64, 62, 67, 68, 62, 64, 65$; $n = 10$.
1. Add; divide by how many:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{645}{10} = 64.5$$

2. Subtract that mean from every value; square:

$$d_i = x_i - \bar{x}, \qquad d_i^2 = (x_i - \bar{x})^2$$

x_i	65	69	59	64	62
$x_i - 64.5$	.5	4.5	−5.5	−.5	−2.5
square	.25	20.25	30.25	.25	6.25

x_i	67	68	62	64	65
$x_i - 64.5$	2.5	3.5	−2.5	−.5	.5
square	6.25	12.25	6.25	.25	.25

3. Add squares; divide by one fewer subject:

$$s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n - 1}$$

$$\sum (x_i - \bar{x})^2 = 82.5, \quad s^2 = \frac{82.5}{10 - 1} = 9.1667$$

4. Square-root for SD; divide SD for SEM:

$$s = \sqrt{s^2}$$

$$s = \sqrt{9.1667} = 3.0277 \text{ in.}$$

$$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{3.0277}{\sqrt{10}} = .95743 \text{ in.}$$

Answer: estimated population mean 64.5 inches; individual SD about 3.03 inches; uncertainty in the estimated mean about .96 inch. **Short calculation for the squared sum:**

$$\sum x_i^2 - \frac{(\sum x_i)^2}{n} = 41685 - \frac{645^2}{10} = 82.5$$

Entire population measured? Use N (all individuals) and divide squared distances by N, not $n - 1$:

$$\mu = \frac{\sum x_i}{N}, \qquad \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Median of these ten heights: sorted middle values 64 and 65; their average is 64.5 inches.

Asked for a sample percentile? Sort, rank, interpolate

Use the lecture rule. p (percent, such as 30), n (sample size); I (integer part of rank), d (decimal part):

$$R = 1 + \frac{p(n-1)}{100} = I + d$$

$$x_p = x_{(I)} + d\big(x_{(I+1)} - x_{(I)}\big)$$

Example: sorted $3, 7, 8, 9, 14$, find 30th percentile.

$$R = 1 + \frac{30(5-1)}{100} = 2.2$$

$$x_{30} = 7 + .2(8 - 7) = 7.2$$

Median: odd n: middle observation; even n: average two middle observations. **IQR:** 75th minus 25th percentile. **Range:** report minimum/maximum; width is maximum minus minimum. Whole rank? Take that observation. Use another convention only if specified.

A probability table → validity, moments, CDF

Recognize: separate possible values with probabilities $f(x)$. **Check first:** every probability nonnegative; all masses sum to one. For .4, .1, .4, total .9: **invalid**. For the valid example:

$$f(x) \geq 0, \qquad \sum_x f(x) = 1$$

x	−1	2	3	7
$f(x)$	.55	.15	.10	.20
$xf(x)$	−.55	.30	.30	1.40
$x^2 f(x)$	.55	.60	.90	9.80

1. Mean: multiply each value by its probability; add.

$$\mu = E[X] = \sum_x x f(x)$$

$$\mu = E[X] = -.55 + .30 + .30 + 1.40 = 1.45$$

2. Mean square: square values, then weight; add.

$$E[X^2] = \sum_x x^2 f(x)$$

$$E[X^2] = .55 + .60 + .90 + 9.80 = 11.85$$

3. Variance: mean square minus squared mean.

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = 11.85 - (1.45)^2 = 9.7475$$

Any requested function: calculate $h(x)$ for each value, multiply by its probability, and add:

$$E[h(X)] = \sum_x h(x) f(x)$$

4. CDF: add every mass at or below the cutoff.

$$F(t) = P(X \leq t) = \sum_{x \leq t} f(x)$$

$$F(x) = \begin{cases} 0 & x < -1 \\ .55 & -1 \leq x < 2 \\ .70 & 2 \leq x < 3 \\ .80 & 3 \leq x < 7 \\ 1 & x \geq 7 \end{cases}$$

Sketch: probability masses at allowed values only; CDF flat between values, jumping at each mass. Include the jump at its actual value. CDF stays between zero and one and never decreases.

If the question asks for squared-distance expectation

Find $E[(X - \mu)^2]$ using $\mu = 1.45$. Subtract this mean from each value, square, then weight by its probability:

$$E[(X-\mu)^2] = \sum_x (x - \mu)^2 f(x)$$

$$= .55(-1 - 1.45)^2 + .15(2 - 1.45)^2$$
$$+ .10(3 - 1.45)^2 + .20(7 - 1.45)^2$$
$$= 3.301375 + .045375 + .24025 + 6.1605$$
$$= 9.7475 = E[X^2] - (E[X])^2$$

Both variance methods agree.

Discrete inequalities: which values enter?

For the table above, $F(2) = .70$, but $P(X < 2) = .55$. The equality includes the mass at 2.

$$P(a \leq X < b) = F(b^-) - F(a^-)$$

$$P(2 \leq X < 7) = .15 + .10 = .25$$

$$P(X > a) = 1 - F(a)$$

$$P(X > 2) = 1 - F(2) = .30$$

$$P(X \geq a) = 1 - F(a^-)$$

$$P(X \geq 2) = 1 - P(X < 2) = .45$$

General: $F(a^-)$ means the accumulated mass strictly below a.

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(a \leq X \leq b) = F(b) - F(a^-)$$

Coins/cards → count paths, then combine chances

Three fair tosses, exactly two heads: HHT, HTH, THH are three of eight equally likely sequences.

$$P(A) = \frac{\text{favorable equally likely outcomes}}{\text{all equally likely outcomes}}$$

$$P(X = 2) = \frac{3}{8}$$

Two kings without replacement: after the first king, three kings remain among fifty-one cards.

$$P(A \cap B) = P(A)P(B \mid A)$$

$$P(\text{two kings}) = \frac{4}{52}\frac{3}{51} = .0045249$$

$$P(A \cap B) = P(A)P(B \mid A)$$

Independent? The first result leaves the second chance unchanged; multiply original probabilities. **Dependent?** Update what remains before multiplying. Three previous heads do not alter the next fair toss: its heads probability stays one half.

Given a density → area, CDF, mean, variance, median

Example: analysis time has $f(x) = x/2$ on $0 \leq x \leq 2$, zero elsewhere. x (hours); $f(x)$ (density height). **1. Check total area:**

$$\int_a^b f(x)\,dx = 1$$

$$\int x^k\,dx = \frac{x^{k+1}}{k+1} + C \quad (k \neq -1)$$

$$\int_0^2 \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_0^2 = 1$$

2. CDF at a cutoff: accumulate from the start to x. Use u as the variable inside the integral:

$$F(t) = \int_a^t f(u)\,du \quad (a \leq t \leq b)$$

$$F(x) = \int_0^x \frac{u}{2}\,du = \left[\frac{u^2}{4}\right]_0^x = \frac{x^2}{4}$$

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2/4 & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

3. Left probability; right probability:

$$P(X < t) = F(t) \quad \text{(continuous)}$$

$$P(X < 1) = F(1) = \frac{1^2}{4} = .25$$

$$P(X > t) = 1 - F(t)$$

$$P(X > 1.5) = 1 - F(1.5) = 1 - \frac{1.5^2}{4} = .4375$$

4. Mean: multiply the density by the value first.

$$\mu = E[X] = \int_a^b x f(x)\,dx$$

$$\mu = \int_0^2 x\frac{x}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{4}{3}$$

5. Mean square; subtract squared mean for variance.

$$E[X^2] = \int_a^b x^2 f(x)\,dx$$

$$E[X^2] = \int_0^2 x^2\frac{x}{2}\,dx = \left[\frac{x^4}{8}\right]_0^2 = 2$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = 2 - (4/3)^2 = 2/9$$

6. Median: solve for half the accumulated area.

$$F(q) = \frac{1}{2}$$

$$F(q) = .5 \;\Rightarrow\; q^2/4 = .5 \;\Rightarrow\; q = \sqrt{2}$$

Answer: mean time 4/3 hours; median about 1.414 hours. The median is not necessarily halfway across the allowed range. Any exact point has probability zero. Density may exceed one, but probability area cannot. Sketch: rising density line; curved CDF that levels at one.

Density changes formula? Accumulate earlier pieces too

Example: triangular density is x from 0 to 1, then $2 - x$ from 1 to 2. The CDF beyond 1 must include the whole first triangle:

$$F(1) = \int_0^1 u\,du = 1/2$$

$$F(x) = \int_a^s f_1(u)\,du + \int_s^x f_2(u)\,du$$

$$F(x) = \frac{1}{2} + \int_1^x (2 - u)\,du$$

$$= \frac{1}{2} + \left[2u - \frac{u^2}{2}\right]_1^x = 2x - \frac{x^2}{2} - 1$$

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2/2 & 0 \leq x \leq 1 \\ 2x - x^2/2 - 1 & 1 < x \leq 2 \\ 1 & x > 2 \end{cases}$$

Check: both middle pieces meet at one half; the end reaches one. For an interval crossing 1, split the density integral there or subtract two complete CDF values.

Shifts, rescaling, and independent sums

Glucose shifted by 26: $E[X - 26] = 95 - 26 = 69$; SD stays 8.7. Adding a constant shifts only the mean. Multiplying scales SD by its magnitude, variance by its square.

$$E[X + c] = \mu + c, \qquad SD(X + c) = \sigma$$

$$E[aX + b] = a\mu + b, \quad \text{Var}(aX + b) = a^2\sigma^2$$

Independent variables: means retain coefficient signs; variances add even for a difference.

$$E[aX + bY + c] = a\mu_X + b\mu_Y + c$$

$$\text{Var}(aX + bY + c) = a^2\sigma_X^2 + b^2\sigma_Y^2$$

A sum of independent Normal variables is Normal.

Unknown density endpoint → make total area one

Given: $f(x) = .1 + x$ for $0 < x \le c$, zero elsewhere. Find c, the unknown upper endpoint. **1. Integrate over the entire allowed range:**

$$\int_a^b f(x)\,dx = 1$$

$$1 = \int_0^c (.1 + x)\,dx = \left[.1x + \frac{x^2}{2}\right]_0^c$$

$$.1c + \frac{c^2}{2} = 1 \;\Rightarrow\; c^2 + .2c - 2 = 0$$

2. Solve the quadratic; keep the positive root:

$$Ac^2 + Bc + D = 0: \quad c = \frac{-B \pm \sqrt{B^2 - 4AD}}{2A}$$

$$c = \frac{-.2 + \sqrt{(.2)^2 - 4(1)(-2)}}{2} = 1.317745$$

3. Accumulate area for the CDF:

$$F(t) = \int_a^t f(u)\,du \quad (a \le t \le b)$$

$$F(x) = \begin{cases} 0 & x \le 0 \\ .1x + x^2/2 & 0 < x \le c \\ 1 & x > c \end{cases}$$

4. Put the requested cutoff into the CDF:

$$P(X > t) = 1 - F(t)$$

$$P(X > .5) = 1 - [.1(.5) + (.5)^2/2] = .825$$

$$P(a < X < b) = F(b) - F(a)$$

$$P(0 < X < .6) = .1(.6) + (.6)^2/2 = .24$$

$$P(X = .5) = 0$$

Sketch: density rises from .1 to its value at the positive endpoint; zero outside. The CDF rises from zero to one and is flat outside the support.

Asked for expected value of a changed measurement?

Use the same density and the unrounded positive endpoint c.

Mean: multiply by x, integrate, then insert c.

$$\mu = E[X] = \int_a^b x f(x)\,dx$$

$$E[X] = \int_0^c x(.1 + x)\,dx$$

$$= \left[.05x^2 + \frac{x^3}{3}\right]_0^c = .05c^2 + \frac{c^3}{3}$$

$$\mu = .849556$$

Now apply the requested linear change:

$$E[aX + b] = aE[X] + b$$

$$E[2X + 1] = 2E[X] + 1 = 2(.849556) + 1 = 2.69911$$

Nonlinear change? Transform each value inside the weighted integral:

$$E[h(X)] = \int h(x) f(x)\,dx$$

Integration reminder: integrate each term, evaluate at the upper limit, then subtract its lower-limit value.

$$\int x^k\,dx = \frac{x^{k+1}}{k+1} + C \quad (k \ne -1)$$

Uniform over a range → requested width / full width

Given bounds: a (start), b (end). The density is flat, with total rectangle area one:

$$f(x) = \frac{1}{b-a}, \quad F(x) = \frac{x-a}{b-a} \quad (a < x \le b)$$

$$P(l < X < u) = \frac{u-l}{b-a} \quad (a \le l < u \le b)$$

Standard Uniform example, zero to one:

$$f(x) = 1, \qquad F(x) = \int_0^x 1\,du = x$$

$$\mu = E[X] = \int_a^b x f(x)\,dx$$

$$E[X] = \int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

$$E[X^2] = \int_a^b x^2 f(x)\,dx$$

$$E[X^2] = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$$

Short route for general endpoints:

$$\mu = \frac{a+b}{2}, \qquad \sigma^2 = \frac{(b-a)^2}{12}$$

CDF is zero below the start, one above the end, and a straight rising line inside. Clip a probability interval to the allowed range before using its width.

Normal measurements → standardize, then read area

Given: glucose $X \sim N(95, 8.7^2)$; $\mu = 95$ (mean), $\sigma = 8.7$ (SD). The second Normal parameter is variance. **Convert each cutoff:**

$$z = \frac{x - \mu}{\sigma}, \quad z_{85} = \frac{85 - 95}{8.7} = -1.149425 \approx -1.15$$

$$z_{115} = \frac{115 - 95}{8.7} = 2.298851 \approx 2.30$$

Your z table: repeated pairs of complete z values and left-tail areas. Find 1.15 and 2.30; read adjacent areas:

$$\Phi(1.15) = .8749, \qquad \Phi(2.30) = .9893$$

Below 85: negative z needs symmetry.

$$\Phi(-z) = 1 - \Phi(z)$$

$$P(X < 85) = \Phi(-1.15) = 1 - .8749 = .1251$$

Between 85 and 115: upper left area minus lower.

$$P(l < X < u) = \Phi(z_u) - \Phi(z_l)$$

$$P(85 \le X < 115) = .9893 - .1251 = .8642$$

At or above 115: complement of the left area.

$$P(X \ge u) = 1 - \Phi(z_u)$$

$$P(X \ge 115) = 1 - .9893 = .0107$$

$$.1251 + .8642 + .0107 = 1$$

Useful checks: $P(X = 95) = 0$; $\sigma^2 = 8.7^2 = 75.69$. Full-precision calculator results are .125190, .864053, .0107567; table rounding explains the small difference.

Below zero glucose?

$$z = (0 - 95)/8.7 = -10.9195$$

The left tail rounds to zero at four decimals; an exact point still has probability zero.

Given a percentile or top-tail area → find the cutoff

97.5th glucose percentile: first interpret the requested area as the probability to the left. Search the z table's area entries for .9750.

$$\Phi(z_p) = p \quad \Longrightarrow \quad z_p = \Phi^{-1}(p)$$

$$\Phi(z_p) = .975 \;\Rightarrow\; z_p = 1.96$$

$$x_p = \mu + z_p\sigma = 95 + 1.96(8.7) = 112.052\ \mathrm{mg/dL}$$

Answer: about 97.5% are below 112.05 mg/dL. **Top 2.5%?** Same cutoff, since left area is 97.5%. **Middle 95%?** Leave .025 in each tail:

$$\mu - 1.96\sigma \quad \text{and} \quad \mu + 1.96\sigma$$

Glucose quartiles: inverse Normal with left areas .25 and .75 gives:

$$z_{.25} = -.67449, \quad z_{.75} = .67449$$

$$Q_1 = 89.1319, \quad Q_3 = 100.8681\ \mathrm{mg/dL}$$

If starting with a right-tail probability, subtract it from one before using a left-area inverse. A percentile above the median should exceed the Normal mean.

If asked to write or sketch a Normal density/CDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

$$F(x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-(u-\mu)^2/(2\sigma^2)}\,du$$

For glucose, insert $\mu = 95, \sigma = 8.7$. Draw a symmetric bell centered at 95; draw an increasing S-shaped CDF with value .5 there, approaching zero and one. The displayed integral is enough for the CDF; it has no elementary antiderivative.

Recognize supplied pnorm and qnorm inputs

pnorm: cutoff → left area. **qnorm:** left area → cutoff. The third argument is SD, not variance. **lower.tail=FALSE** means a right-tail area.

$$P(X \le x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$\texttt{pnorm}(85, 95, 8.7) \approx .125190$$

$$x_p = \mu + z_p\sigma$$

$$\texttt{qnorm}(.975, 95, 8.7) \approx 112.0517$$

For a sample-mean distribution, replace the SD input with SEM. R is the calculation engine; RStudio is its interface. A vector stores values; a data frame stores variables by column and observations by row. Interpret supplied commands and output; code writing is not the exam task.

The question says "average of n" → replace SD with SEM

Given impurity: individual mean $\mu = 4$, individual SD $\sigma = 1.5$ grams; $n = 50$ independent batches. Find $P(3.5 < \bar{X} < 3.8)$. **1. The average keeps the mean but gets a smaller spread:**

$$\mathrm{SEM} = \frac{\sigma}{\sqrt{n}}$$

$$E[\bar{X}] = \mu = 4, \qquad \mathrm{SEM} = \frac{1.5}{\sqrt{50}} = .212132$$

2. Standardize both bounds using that SEM:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$z_L = \frac{3.5 - 4}{.212132} = -2.35702$$

$$z_U = \frac{3.8 - 4}{.212132} = -.942809$$

3. Subtract accumulated probabilities:

$$P(l < X < u) = \Phi(z_u) - \Phi(z_l)$$

$$P = \Phi(-.942809) - \Phi(-2.35702)$$

$$= .172889 - .00921106 = .163678$$

Using your table: round to −.94 and −2.36; use symmetry, then subtract. The rounded-table answer is .1645. **Answer:** about 16.4% of such sample averages fall in the requested range.

Know which spread the question needs

One individual: use σ. **Mean of n:** use $\sigma/\sqrt{n}$. **Unknown population SD:** estimate with s. Individual SD describes people; SEM describes repeated sample means.

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Normal individuals give a Normal mean at any sample size. Otherwise the course's CLT guideline uses sufficiently large independent samples, around thirty or more. The original individuals do not become Normal.

Why the square root? Independent variances add, and averaging scales variance by the square of its multiplier:

$$\mathrm{Var}\left(\frac{\sum X_i}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Four times the subjects halves SEM. Unbiased means correctly centered across repeated samples. For a Normal population, the sample mean is the minimum-variance center estimator discussed in class. If asked for uncertainty in variance:

$$\widehat{\mathrm{SE}}(s^2) = s^2\sqrt{\frac{2}{n-1}}$$

TI-84: enter these when the arithmetic needs help

Normal interval: 2nd → VARS (DISTR) → normalcdf. Enter lower=85, upper=115, mean=95, SD=8.7 → Calculate/Paste then ENTER. This gives .864053. **Mean impurity:** lower=3.5, upper=3.8, mean=4, SD=1.5/√50. The last input is SEM.

Percentile: DISTR → invNorm: left area=.975, mean=95, SD=8.7 → 112.0517. For an open tail, use −1E99 or 1E99 as the practical bound. E is entered with 2nd → comma (EE); use the (-) key for a negative entry.

Raw data: STAT → EDIT → enter list in L1; STAT → CALC → 1-Var Stats. Read x-bar, **Sx (sample SD)**, n. σx divides by the list size; use Sx for a sample. To clear a list: highlight its name, CLEAR, ENTER.

Keep intermediate values: STO→ A after an intermediate; use A in the next calculation. Write A's meaning beside it. Put an entire denominator in parentheses; keep extra digits until the final answer.

Data type and shape: what to say

Continuous: height/BP. **Discrete:** counts. **Nominal:** country/ZIP/event. **Ordinal:** ordered stage/pain.

Histogram: bin counts. **Box:** Q1–Q3 with median; IQR=Q3−Q1; whisker rule uses 1.5×IQR. **Violin:** width shows concentration. **Q–Q:** roughly straight supports Normal shape; curvature suggests mismatch. Normal: 68%, 95%, 99.7% within one, two, three SDs.

ONE sample vs a reference → one-sample t test

Question: is mean student HR different from 72 BPM? **Given:** $\bar{x} = 76.8$ (sample mean), $s = 4.15$ (sample SD), $n = 5$, $\mu_0 = 72$ (reference), $\alpha = .05$.

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0$$
$$H_0 : \mu = 72, \qquad H_1 : \mu \neq 72$$

1. Mean uncertainty; 2. observed discrepancy:

$$\text{SEM} = s/\sqrt{n} = 4.15/\sqrt{5} = 1.855936$$
$$\bar{x} - \mu_0 = 76.8 - 72 = 4.8$$

3. Divide discrepancy by its uncertainty:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{4.8}{1.855936} = 2.58630$$

4. Table: sample size minus one gives degrees of freedom; your two-tailed t table uses the full .05 column.

$$\nu = n - 1 = 4, \quad t_{\text{crit}} = 2.776$$
$$|t| > t_{\text{crit}} \;\Rightarrow\; \text{reject } H_0$$
$$|2.58630| < 2.776 \;\Rightarrow\; \text{do not reject } H_0$$

Write: "Using a two-sided one-sample t test at .05, these data do not establish that the population mean student heart rate differs from 72 BPM." **Shortcut:** difference from reference ÷ SEM. The fixed reference is not another sampled group.

TWO independent means → pooled t test

Question: do control and intervention populations differ in cognitive score? $(\bar{x}_1, s_1, n_1) = (49, 8.4, 22)$, $(\bar{x}_2, s_2, n_2) = (57, 9.5, 27)$; entries are **mean, SD, count**.

$$H_0 : \mu_1 = \mu_2, \qquad H_1 : \mu_1 \neq \mu_2$$

1. Square SDs, then weight by count minus one:

$$s_1^2 = 8.4^2 = 70.56, \quad s_2^2 = 9.5^2 = 90.25$$
$$\nu = n_1 + n_2 - 2 = 22 + 27 - 2 = 47$$
$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$= \frac{21(70.56) + 26(90.25)}{47} = 81.45234$$

2. Standard error of the difference:

$$\text{SE}_\Delta = \sqrt{s_p^2(1/n_1 + 1/n_2)}$$
$$= \sqrt{81.45234(1/22 + 1/27)} = 2.592129$$

3. Difference divided by that standard error:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}_\Delta} = \frac{49 - 57}{2.592129} = -3.08627$$

4. Table: df 47 is between rows 46 and 48; the .05 two-tailed critical magnitude lies between 2.013 and 2.011. The observed magnitude exceeds both.

$$|t| > t_{\text{crit}} \;\Rightarrow\; \text{reject } H_0$$
$$|t| = 3.08627 > 2.013 \;\Rightarrow\; \text{reject } H_0$$

Write: "At .05, a pooled two-sample t test supports different population mean cognitive scores; the intervention group's sample mean is higher." **Equal n only:** pooled variance simplifies to the average of the two variances. **Unequal n:** retain the count-minus-one weights.

Given t output? Start from its statistic, df, p

Identify: one-sample reference or two independent groups. Read the reported t, df, p, means, and alternative. For a two-sided test, compare p with the stated alpha, or |t| with the two-tailed critical value.

If asked to show the setup: state the population null, identify the numerator difference, and identify the appropriate SE. A negative t describes group order, not a negative probability. "Do not reject" means a difference was not established, not that equality was proved.

TI-84 checks for the two worked mean tests

One sample: STAT → TESTS → T-Test → Stats; reference=72, mean=76.8, Sx=4.15, n=5; choose ≠; Calculate.

Two samples: STAT → TESTS → 2-SampTTest → Stats; mean1=49, Sx1=8.4, n1=22; mean2=57, Sx2=9.5, n2=27; choose ≠; **Pooled: Yes**; Calculate.

SEVERAL equal-size groups → balanced ANOVA

Question: do population mean tumor sizes differ across three dosing groups? $m = 3$ (groups), $n = 12$ (subjects **in each**); means $1188, 1412, 1363$; SDs $189, 217, 191$.

$$H_0 : \mu_1 = \mu_2 = \mu_3; \quad H_1 : \text{not all means equal}$$

1. Within variance: square the SDs; average.

$$s_{\text{wit}}^2 = \frac{\sum_{i=1}^{m} s_i^2}{m}$$
$$s_{\text{wit}}^2 = \frac{189^2 + 217^2 + 191^2}{3}$$
$$= \frac{35721 + 47089 + 36481}{3} = 39763.6667$$

2. Grand mean: average the group means.

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{m} \bar{x}_i}{m}$$
$$\bar{\bar{x}} = \frac{1188 + 1412 + 1363}{3} = 1321$$

3. Distances of group means from grand mean:

$$d_i = \bar{x}_i - \bar{\bar{x}}$$
$$1188 - 1321 = -133, \quad 1412 - 1321 = 91$$
$$1363 - 1321 = 42$$
$$\sum(\bar{x}_i - \bar{\bar{x}})^2 = (-133)^2 + 91^2 + 42^2 = 27734$$

4. Variance of means; rescale by subjects per group:

$$s_{\bar{x}}^2 = \frac{\sum_i (\bar{x}_i - \bar{\bar{x}})^2}{m - 1}$$
$$s_{\bar{x}}^2 = \frac{27734}{m - 1} = \frac{27734}{2} = 13867$$
$$s_{\text{bet}}^2 = n s_{\bar{x}}^2 = 12(13867) = 166404$$

5. Between ÷ within; 6. degrees of freedom:

$$F = \frac{s_{\text{bet}}^2}{s_{\text{wit}}^2} = \frac{166404}{39763.6667} = 4.18483$$
$$\nu_n = m - 1 = 2, \qquad \nu_d = m(n - 1) = 33$$

7. Your F table: numerator df 2 across the top; denominator df down the side. For .05 read **lightface**. Row 32 gives 3.29; row 34 gives 3.28. The df-33 cutoff lies between them.

$$F > F_{\text{crit}} \;\Rightarrow\; \text{reject } H_0$$
$$4.18483 > 3.29 \;\Rightarrow\; \text{reject } H_0$$

Write: "At .05, one-way ANOVA supports a difference in population mean tumor size among the dosing groups." It does not identify the differing pairs. **Memory route:** average variances → variance of means × n → divide.

Supplied ANOVA output → use the Residuals row

$$F = \frac{\text{MS}_{\text{Group}}}{\text{MS}_{\text{Residual}}}$$

row	df	Mean Sq	F
Group	2	389.7	5.832
Residuals	19	66.8	

Given p=.0106: compare with alpha. At .05 reject equal population means; at .01 do not reject. The printed F is the ratio of the two mean squares, allowing for rounded entries.

Continue with pairwise tests? Use $s_{\text{wit}}^2 = 66.8$ from Residual Mean Sq and $\nu = 19$ from Residual df. Keep the actual pair's sample sizes in its SE. Do not use residual standard error as though it were variance; square it first if it is the only spread supplied.

Conditions and decisions to name when asked

Mean tests: numerical interval outcome, independent randomly selected observations, Normal population(s). **Pooled t and ANOVA:** also common population variance. Balanced ANOVA uses equal group sizes. A Normal Q–Q plot should be approximately straight. **t:** compare $|t|$ with its positive cutoff for a two-sided alternative. **F and χ²:** upper-tail cutoff. Or compare the reported p with alpha. A p value is an extreme-result probability **under the null**, not the chance the null is true.

If ANOVA does not cross its cutoff

Cat/dog heart rate: two groups of five.

$$F = \frac{s_{\text{bet}}^2}{s_{\text{wit}}^2}$$
$$F = 16.9/68.05 = .248, \quad (\nu_n, \nu_d) = (1, 8)$$
$$F > F_{\text{crit}} \;\Rightarrow\; \text{reject } H_0$$
$$.248 < 5.32 \;\Rightarrow\; \text{do not reject } H_0$$

Write: "At .05, a difference in population mean heart rate was not established."

After significant ANOVA → which specific means differ?

Use the ANOVA's within variance and residual df. For each requested pair i, j:

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{\sqrt{s_{\text{wit}}^2(1/n_i + 1/n_j)}}$$

Equal-size shortcut: denominator is $\sqrt{2s_{\text{wit}}^2/n}$. **Count only the requested comparisons:** all pairs use $m(m-1)/2$; each treatment against one control uses $m - 1$.

Bonferroni worked: three filgrastim dose groups

Given ANOVA: means $3.7, 4.6, 5.2$, $n = 15$ each, within variance 2.5267, residual df 42; compare all three pairs at overall $\alpha_T = .05$.

$$\text{SE}_{ij} = \sqrt{\frac{2s_{\text{wit}}^2}{n}} \quad (n_i = n_j = n)$$
$$\text{SE} = \sqrt{2(2.5267)/15} = .580425$$
$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{\text{SE}_{ij}}$$
$$|t_{13}| = \frac{5.2 - 3.7}{.580425} = 2.58431$$
$$|t_{12}| = \frac{4.6 - 3.7}{.580425} = 1.55059$$
$$|t_{23}| = \frac{5.2 - 4.6}{.580425} = 1.03373$$

One adjusted level for each of k requested tests:

$$\alpha_{\text{pair}} = \alpha_T/k = .05/3 = .016667$$

Missing table column? TI-84: DISTR → invT; left area=1−.05/(2×3), df=42 → 2.49367. This halves the tail for the inverse calculation, not for your two-tailed table's column.

$$P(T_\nu > t_{\text{crit}}) = \frac{\alpha_T}{2k}$$
$$2.58431 > 2.49367; \quad 1.55059, 1.03373 < 2.49367$$

Write: "With Bonferroni control at overall .05, low and high dose population means differ; low–medium and medium–high differences were not established."

Holm worked: strongest first; stop at the first failure

Given: five comparisons; residual df 46; overall $\alpha_T = .05$. Sort absolute t values largest first, keeping their group names:

j	pair	$\lvert t\rvert$	α_j
1	1 : 6	3.3	.0100
2	1 : 3	2.9	.0125
3	1 : 5	2.3	.016667
4	1 : 4	2.2	.0250
5	1 : 2	2.1	.0500

$$\alpha_j = \frac{\alpha_T}{k - j + 1}$$

Find each positive critical t using df 46:

$$3.3 > 2.687 \;\Rightarrow\; \text{reject; continue}$$
$$2.9 > 2.59973 \;\Rightarrow\; \text{reject; continue}$$
$$2.3 < 2.48469 \;\Rightarrow\; \text{stop}$$

Write: "At familywise .05, G1 differs from G6 and G3; differences from G5, G4, and G2 were not established." Do not restart at the easier later cutoffs. With supplied p values, sort p smallest first and compare directly with each adjusted level.

t-table / TI-84 lookup without mixing the tails

Your t table: row is df; column is **total two-tail** significance. At df 4, column .05 gives 2.776. Do not halve alpha again. For a two-sided Normal z cutoff, the infinity-df row gives 1.960 at .05.

TI-84 positive t critical: DISTR → invT; enter left area $1 - \alpha/2$, then df. For Holm, repeat using each α_j. If invT is unavailable, supplied p values or table bounds may decide; do not invent an exact missing cell.

Bounds: a statistic above both bracketing cutoffs rejects; below both does not. If it lies between them, the bounds alone do not settle the test. Use an exact calculator cutoff or supplied p. **Two-sided t p:** DISTR → tcdf, lower=|t|, upper=1E99, df; multiply that right-tail area by two.

One event proportion → estimate and standard error

Example: death in 8 of 61 patients. Code death=1, survival=0; e (events), n (total).

$$\hat{p} = e/n = 8/61 = .131148$$
$$\widehat{\mathrm{Var}}(X) = \hat{p}(1-\hat{p}) = .113948$$
$$\widehat{\mathrm{SD}}(X) = \sqrt{\hat{p}(1-\hat{p})}$$
$$\widehat{\mathrm{SD}}(X) = \sqrt{.113948} = .337562$$
$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{.113948}{61}} = .0432203$$

The first spread describes individual zero-one outcomes; the last describes uncertainty in the estimated proportion. Use the same named event in every compared group.

TWO event proportions → Yates-corrected z

Question: does aspirin change thrombus occurrence? Placebo: 18/25; aspirin: 6/19. $n_1 = 25, n_2 = 19$ (group totals), $e_1 = 18, e_2 = 6$ (events).

$$H_0 : p_1 = p_2, \qquad H_1 : p_1 \neq p_2$$

1. Group rates; pool actual events, not rates:

$$\hat{p}_i = \frac{e_i}{n_i}$$
$$\hat{p}_1 = 18/25 = .72, \qquad \hat{p}_2 = 6/19 = .315789$$
$$\hat{p} = \frac{e_1 + e_2}{n_1 + n_2} = \frac{24}{44} = .545455$$

2. Build the pooled null standard error:

$$\mathrm{SE}_0 = \sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$$
$$= \sqrt{.545455(.454545)(1/25 + 1/19)}$$
$$= \sqrt{.0229665} = .151547$$

3. Reduce the absolute gap before dividing:

$$c_Y = \frac{1}{2}(1/25 + 1/19) = .0463158$$
$$D_Y = \max(0, |\hat{p}_1 - \hat{p}_2| - c_Y)$$
$$D_Y = \max(0, |.72 - .315789| - .0463158) = .357895$$
$$|z_Y| = D_Y/\mathrm{SE}_0 = .357895/.151547 = 2.36161$$

4. Your two-tailed t table: infinity-df row, .05 column → 1.960.

$$|z| > z_{\text{crit}} \;\Rightarrow\; \text{reject } H_0$$
$$2.36161 > 1.960 \;\Rightarrow\; \text{reject } H_0$$

Write: "At .05, the corrected two-proportion z test supports different population thrombus rates; the aspirin group's observed rate is lower." **Use Yates for 2×2 tests.** If given rounded proportions and an exact total event count, pool that exact count.

Same 2×2 counts → corrected χ², cell by cell

Rows: placebo, aspirin. **Columns:** thrombus, none. Observed rows $(18, 7), (6, 13)$; row totals $25, 19$, column totals $24, 20$, grand total $N = 44$. **H₀:** same event proportions.

$$E_{ij} = \frac{R_i C_j}{N} \quad (\text{expected cell count})$$
$$E = \begin{pmatrix} 25(24)/44 & 25(20)/44 \\ 19(24)/44 & 19(20)/44 \end{pmatrix}$$
$$= \begin{pmatrix} 13.63636 & 11.36364 \\ 10.36364 & 8.63636 \end{pmatrix}$$

Each cell: absolute observed–expected gap, subtract .5, square, divide by its expected count.

$$c_{ij} = \frac{\max(0, |O_{ij} - E_{ij}| - .5)^2}{E_{ij}}$$
$$\frac{(|18 - 13.63636| - .5)^2}{13.63636} = 1.094697$$
$$\chi^2_Y = \sum \frac{\max(0, |O - E| - .5)^2}{E}$$
$$= 1.094697 + 1.313636 + 1.440391 + 1.728469$$
$$\nu = (r-1)(c-1)$$
$$= 5.577193, \qquad \nu = (2-1)(2-1) = 1$$

χ² table: row 1, upper-tail .05 column → 3.841. Since 5.577193 exceeds it, reject; aspirin and placebo event rates differ. Check: corrected z squared gives the same χ². Standard calculator routines may omit Yates; retain the corrected written setup.

If asked for a p range: at df 1, 5.577 lies between 5.024 (.025 tail) and 6.635 (.01 tail).

$$.01 < p < .025$$

MORE rows/columns → ordinary contingency χ²

Three treatments, two outcomes: observed rows $(36, 131), (41, 132), (75, 104)$. Row totals $167, 173, 179$; column totals $152, 367$; $N = 519$. **H₀:** equal population outcome distributions across treatments.

1. Expected counts: use this table's margins.

$$E_{ij} = \frac{R_i C_j}{N}$$
$$E_{11} = \frac{167(152)}{519} = 48.90944$$
$$E = \begin{pmatrix} 48.90944 & 118.09056 \\ 50.66667 & 122.33333 \\ 52.42389 & 126.57611 \end{pmatrix}$$

2. No Yates for this 3×2 table. First cell:

$$c_{ij} = \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$\frac{(36 - 48.90944)^2}{48.90944} = 3.407393$$

3. Do every cell the same way and add:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$= 3.407393 + 1.411236 + 1.844298 + .763851$$
$$+9.722297 + 4.026673 = 21.175749$$
$$\nu = (r-1)(c-1) = (3-1)(2-1) = 2$$

4. χ² table: row 2, upper-tail .05 → 5.991. Reject because 21.175749 exceeds it. **Write:** "The population outcome distribution differs among the treatment groups." For follow-ups, create each requested 2×2 subtable, recompute its margins, use Yates, then adjust the family. Expected cells may be fractional. Their row/column totals must match the observed margins. Use counts, not percentages, in the statistic.

ONE list vs specified probabilities → goodness of fit

Exercise categories: observed $255, 125, 90$; total $N = 470$. Null target probabilities $.60, .25, .15$. **H₀:** each population category probability equals its stated target.

1. Expected = total × target probability:

$$E_i = N p_{i,0}$$
$$E = (470(.60), 470(.25), 470(.15)) = (282, 117.5, 70.5)$$

2. Subtract, square, divide, and add:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$
$$\chi^2 = \frac{(255 - 282)^2}{282} + \frac{(125 - 117.5)^2}{117.5}$$
$$+ \frac{(90 - 70.5)^2}{70.5}$$
$$= 2.585106 + .478723 + 5.393617 = 8.457447$$
$$\nu = k - 1 = 3 - 1 = 2$$

3. χ² table: row 2, upper-tail .05 → 5.991. Reject. **Write:** "The current exercise-category distribution differs from the specified earlier pattern." Do not derive expected probabilities from the current counts; the null supplies them.

Given pairwise χ² output → requested pairs + Holm

Only control comparisons requested: control–G1: 9.4944; control–G2: 4.6895; control–G3: 18.234. Each is corrected 2×2 with df 1. Although six pairs were printed, this family has $k = 3$.

1. Sort largest first: G3, G1, G2. **2. Use overall .05 and successive levels:**

$$\alpha_j = \frac{\alpha_T}{k - j + 1}$$
$$\alpha_j = .05/3, \quad .05/2, \quad .05$$

3. Read row 1 of your χ² table:

$$18.234 > 6.635 > \chi^2_{.016667,1} \;\Rightarrow\; \text{reject}$$
$$9.4944 > \chi^2_{.025,1} = 5.024 \;\Rightarrow\; \text{reject}$$
$$4.6895 > \chi^2_{.05,1} = 3.841 \;\Rightarrow\; \text{reject}$$

First step: the .016667 column is absent, but the stricter .01 cutoff is 6.635 and is already exceeded. **Write:** "At overall .05, each of G1, G2, and G3 differs from control." Keep the named outcome in the final sentence. For any first failure, stop.

Prospective risks → RR, absolute reduction, NNT

Event: thrombi. **Treatment:** aspirin 6/19; **reference:** placebo 18/25. A trial/cohort supplies actual group-risk denominators.

$$r_T = \frac{e_T}{n_T}, \qquad r_C = \frac{e_C}{n_C}$$
$$r_T = 6/19 = .315789, \qquad r_C = 18/25 = .72$$

1. Relative risk: divide risks.

$$\mathrm{RR} = \frac{r_T}{r_C} = \frac{6/19}{18/25} = .438596$$

Say: aspirin's thrombus risk is about .44 times placebo's.
2. Absolute reduction: subtract risks.

$$\mathrm{ARR} = r_C - r_T = .72 - .315789 = .404211$$

Say: about 40.42 percentage points lower event risk. **3. NNT: reciprocal of that absolute reduction.**

$$\mathrm{NNT} = 1/\mathrm{ARR} = 1/.404211 = 2.47396$$

Say: about 2.47 treated patients per additional avoided thrombus; if a whole-person answer is requested, round up to **3**. Zero absolute change gives no finite NNT.

Case-control study → exposure odds ratio

Cases: 50 exposed, 14 unexposed. **Controls:** 43 exposed, 35 unexposed. Selection began with disease status. **1. Exposure odds within each outcome group:**

$$\mathrm{odds}_1 = \frac{a}{b}, \qquad \mathrm{odds}_0 = \frac{c}{d}$$
$$\text{case odds} = 50/14, \qquad \text{control odds} = 43/35$$

2. Divide; use the diagonal products shortcut:

$$\mathrm{OR} = \frac{a/b}{c/d} = \frac{ad}{bc}$$
$$\mathrm{OR} = \frac{50/14}{43/35} = \frac{50(35)}{14(43)} = 2.90698$$

Write: "The odds of secondhand-smoke exposure are about 2.91 times higher among cases than controls." Case-control totals do not directly estimate population disease risks. **Generic table:**

	event	none
exposed	a	b
unexposed	c	d

$$\mathrm{OR} = \frac{ad}{bc}$$

Risk = events/all; odds = events/non-events. Reverse one comparison → reciprocal. RR/OR=1 means no association.

Benefit vs harm: use absolute probabilities

Given: death risk falls from 10 per thousand to 8 per thousand. A "20% reduction" is relative; start by converting counts to risks.

$$\mathrm{RR} = \frac{r_T}{r_C}$$
$$r_C = .010, \quad r_T = .008, \quad \mathrm{RR} = .8$$
$$\mathrm{RRR} = 1 - \mathrm{RR}$$
$$\mathrm{RRR} = 1 - .8 = .20 \quad (20\%)$$
$$\mathrm{ARR} = r_C - r_T$$
$$\mathrm{ARR} = .010 - .008 = .002 \quad (.2 \text{ percentage points})$$
$$\mathrm{NNT} = \frac{1}{\mathrm{ARR}}$$
$$\mathrm{NNT} = 1/.002 = 500$$

Harm increases? Use the absolute increase for NNH:

$$\mathrm{NNH} = \frac{1}{r_T - r_C} \quad (r_T > r_C)$$

If NNH=20: treating five hundred gives twenty-five additional harms per additional avoided death.

$$\text{additional harms per benefit} = \frac{\mathrm{NNT}}{\mathrm{NNH}}$$
$$500/20 = 25$$

Lower NNT means a more frequent additional benefit; lower NNH means a more frequent additional harm. Use decimal probabilities in reciprocals; 20% and twenty percentage points are different.

Study-design questions: identify what actually happened

Trial: random assignment then follow-up. **Cohort:** follow exposures without randomization. **Case-control:** select outcomes, assess earlier exposures. **RR:** prospective risks; **OR:** case-control odds.

Randomization: group comparability. **Masking:** expectation bias; single=patient, double=also care team, triple=also statistical team. **Sham:** procedure-experience control. **Parallel:** different people. **Crossover:** same people, different periods; stable disease, no carryover. Prespecify primary question; monitor recruitment/adherence. Volunteers may differ from other patients. Association alone is not causation.